

ADA Global Academy

MicroMasters in Data Science

Comprehensive Machine Learning Capstone Assignment

Machine Learning with Python, R, and JMP

Assignment Type: Individual Capstone Project

Duration: 2 Weeks

Submission Format: Report + Runnable Code + Presentation Video

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1 Assignment Overview

This capstone assignment is designed to evaluate students' ability to independently conduct a complete machine learning study using real-world data.

Students are expected to:

- identify and obtain a suitable dataset,
- clean and prepare the data,
- conduct exploratory analysis,
- build machine learning models,
- evaluate model performance,
- interpret findings,
- communicate insights professionally,
- and submit fully reproducible code.

Students may use:

- Python,
- R,
- JMP,
- or a combination of these tools.

The goal is not only to achieve high prediction accuracy, but also to demonstrate sound analytical reasoning, interpretability, and practical understanding of machine learning workflows.

2 Learning Objectives

By completing this project, students should be able to:

1. Source and evaluate real-world datasets
2. Conduct data cleaning and preprocessing
3. Apply supervised or unsupervised machine learning techniques

4. Compare multiple machine learning algorithms
5. Evaluate model performance appropriately
6. Interpret results meaningfully
7. Produce reproducible analytical workflows
8. Communicate findings in professional report format
9. Demonstrate ethical and responsible use of data
10. Present actionable insights from data analysis

3 Dataset Selection

Students must independently source a dataset from:

- Kaggle
- UCI Machine Learning Repository
- WHO
- World Bank
- Government Open Data Portals
- GitHub Repositories
- Public APIs
- Research Repositories
- Other reputable sources

Datasets may relate to:

- Health
- Finance
- Education
- Agriculture
- Climate

- Social Sciences
- Sports
- Transportation
- Business
- Civic Data
- Marketing
- Cybersecurity

3.1 Dataset Conditions

The dataset should:

- contain sufficient observations,
- contain multiple variables/features,
- require preprocessing,
- and be suitable for machine learning analysis.

Students are encouraged to select datasets with practical or societal relevance.

4 Acceptable Project Types

Students may choose ONE of the following:

4.1 Classification Project

Examples:

- Disease prediction
- Customer churn prediction
- Spam detection
- Fraud detection
- Sentiment classification

4.2 Regression Project

Examples:

- House price prediction
- Rainfall forecasting
- Sales prediction
- Stock prediction
- Energy consumption estimation

4.3 Clustering Project

Examples:

- Customer segmentation
- Regional grouping
- Behavioral clustering

4.4 Time Series Forecasting

Examples:

- Stock forecasting
- Weather forecasting
- Epidemic trend forecasting

4.5 Explainable AI Project

Examples:

- SHAP
- LIME
- Interpretable machine learning
- Feature importance analysis

5 Required Project Components

5.1 Section 1: Introduction

Include:

- Background of the problem
- Motivation
- Research objectives
- Significance of the study
- Description of the dataset

5.2 Section 2: Data Collection and Description

Provide:

- Source of the dataset
- Number of observations
- Number of variables
- Variable descriptions
- Target variable (if supervised learning)
- Data limitations

5.3 Section 3: Data Cleaning and Preprocessing

Demonstrate:

- Handling missing values
- Handling duplicates
- Outlier analysis
- Encoding categorical variables
- Scaling/normalization
- Feature engineering
- Train-test splitting

Students should explain why each preprocessing step was necessary.

5.4 Section 4: Exploratory Data Analysis (EDA)

Include:

- Descriptive statistics
- Visualizations
- Correlation analysis
- Distribution plots
- Insights from the data

Possible visualizations:

- Histograms
- Boxplots
- Scatterplots
- Heatmaps
- Pair plots
- Bar charts
- Line charts

Students must interpret the visuals, not just present them.

5.5 Section 5: Machine Learning Modeling

Students must implement at least TWO machine learning models.

5.5.1 Classification Models

Examples:

- Logistic Regression
- Decision Tree
- Random Forest
- XGBoost

- CatBoost
- Support Vector Machine
- KNN
- Naive Bayes
- Neural Networks

5.5.2 Regression Models

Examples:

- Linear Regression
- Random Forest Regressor
- Gradient Boosting
- XGBoost Regressor
- Support Vector Regression

5.5.3 Clustering Models

Examples:

- K-Means
- Hierarchical Clustering
- DBSCAN

5.5.4 Time Series Models

Examples:

- ARIMA
- Prophet
- LSTM
- Exponential Smoothing

6 Model Evaluation

Students must use appropriate evaluation metrics.

6.1 Classification Metrics

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC
- Confusion Matrix

6.2 Regression Metrics

- RMSE
- MAE
- MSE
- R^2

6.3 Clustering Metrics

- Silhouette Score
- Cluster Interpretation

Students must compare models and justify which model performed best.

7 Interpretation and Discussion

Students should discuss:

- what the model learned,
- important predictors/features,
- implications of findings,

- limitations,
- ethical considerations,
- practical applications.

Students are strongly encouraged to use:

- Feature importance
- SHAP values
- LIME explanations
- Partial dependence plots
- Interpretable visualization techniques

8 Conclusion and Recommendations

Include:

- Summary of findings
- Best-performing model
- Recommendations
- Future improvements
- Possible real-world implementation

9 Reproducibility

Students must submit:

- Runnable code
- Datasets (or source links)
- Package requirements
- Instructions for execution

The code should run successfully without major modification.

10 Technical Requirements

Students may use Python, R, JMP, or combinations of these tools.

10.1 Suggested Python Libraries

- pandas
- numpy
- scikit-learn
- matplotlib
- seaborn
- xgboost
- catboost
- shap
- tensorflow
- keras

10.2 Suggested R Packages

- tidyverse
- caret
- randomForest
- xgboost
- DALEX
- ggplot2
- tidymodels

10.3 JMP Requirements

Students using JMP must:

- clearly document workflows,
- include screenshots,
- explain analysis steps,
- and provide exported outputs.

11 Deliverables

11.1 Final Report

- Format: PDF or Word document
- Recommended Length: 10–20 pages
- Must include figures, tables, interpretations, and references

11.2 Runnable Code

Acceptable formats:

- .py
- .ipynb
- .R
- .Rmd
- JMP project files

Code must:

- be organized,
- commented,
- and executable.

11.3 Presentation Slides

Prepare 5–10 presentation slides summarizing:

- problem,
- methods,
- findings,
- conclusions.

12 Suggested Project Topics

Students may choose topics such as:

- Malaria prediction
- Student performance prediction
- Loan default prediction
- Climate trend analysis
- Credit card fraud detection
- Customer segmentation
- Stock market forecasting
- Social media sentiment analysis
- Healthcare risk prediction
- Agricultural yield prediction
- Energy demand forecasting
- Air quality prediction

13 Assessment Rubric

Component	Marks
Problem formulation and dataset selection	10
Data cleaning and preprocessing	15
Exploratory data analysis	15
Machine learning implementation	20
Model evaluation and comparison	15
Interpretation and discussion	15
Code quality and reproducibility	5
Report presentation and organization	5
Total	100

14 Important Instructions

- Plagiarism is strictly prohibited.
- Students must clearly cite dataset sources and references.
- Reports generated entirely by AI without understanding or interpretation are unacceptable.
- Students must be prepared to defend their work orally if requested.
- Originality, interpretation, and analytical reasoning will be strongly rewarded.

15 Bonus Marks Opportunities

Learners may receive bonus marks for:

- Deploying models
- Building dashboards
- Creating web applications
- Advanced explainability
- Novel feature engineering
- Comparing many algorithms
- Using deep learning appropriately

- Integrating civic or societal impact analysis

16 Final Note to Learners

This assignment is intended to simulate a real-world machine learning project from start to finish. Focus on:

- understanding the data,
- building meaningful models,
- interpreting findings clearly,
- and communicating insights professionally.

High-quality interpretation and analytical reasoning are more important than merely obtaining the highest accuracy score.

Best Wishes

Prof. O. O. Awe,
for ADA Global Academy